

Activity-3

1) Given $G_x = 24$

$G_y = 18$

1) Gradient

magnitude : $\sqrt{G_x^2 + G_y^2}$

$$= \sqrt{(24)^2 + (18)^2}$$

$$= \sqrt{576 + 324}$$

$$= \sqrt{900}$$

$$= 30.$$

2) Gradient

direction : $\theta = \tan^{-1} (G_y/G_x)$

$$\theta = \tan^{-1} (18/24)$$

$$\theta = \tan^{-1} (0.75)$$

$$\boxed{\theta = 36.87^\circ}$$

2) Given

Pixel

Magnitude

P₁

25

P₂

50

P₃

90

P₄

60

P₅

30

Since the gradient direction is horizontal.
compare each pixel with its left and right neighbours

$P_1 \rightarrow$ Border pixel $\rightarrow$ Suppressed (0)

$P_2 \rightarrow 50 \Rightarrow$ compare with 25 and 90

No maximum 50(0)

$P_3 \rightarrow 90 \Rightarrow$ compare with 50 and 60.

It is maximum 50(90)

$P_4 \rightarrow 60 \Rightarrow$ compare with 90 and 30

Not max 50(0).

$P_5 \Rightarrow 50 \Rightarrow$ border 50(0) Suppressed.

Pixel	P_1	P_2	P_3	P_4	P_5
output	0	0	90	0	0

3. Given

High threshold = 100

Low threshold = 150

Classification.

pixels & magnitude

135	P_1	$P_1 = 135 > 100$	strong edge
90	P_2	$P_2 = 90 < 100$	weak
45	P_3	$P_3 = 45 < 100$	discard
120	P_4	$P_4 = 120 > 100$	strong
70	P_5	$P_5 = 70 < 100$	weak.
20	P_6	$P_6 = 20 < 100$	discard.

Strong edges: P_1, P_4

Weak edges: P_2, P_5

Non edges: P_3, P_6

4. Given

Pixel Magnitude

P_1

135

P_2

95

P_3

40

P_4

115

P_5

65

P_6

25

P_7

105

HT = 100

LT = 50

Given

P_2, P_5 connected to P_1

P_5 is not connected to any

Strong edges.

Strong

edges: P_1, P_4, P_7

Weak

edges: P_2, P_5

Non

edges: P_3, P_6

Hysteresis:

P_2, P_5
as edge.

P_5 is

removed

Final edge

Hysteresis:

p_2 is connected to strong edge p_1 so keep as edge.

p_5 is not connected to any edge so remove.

Final edge pixels after Hysteresis:

